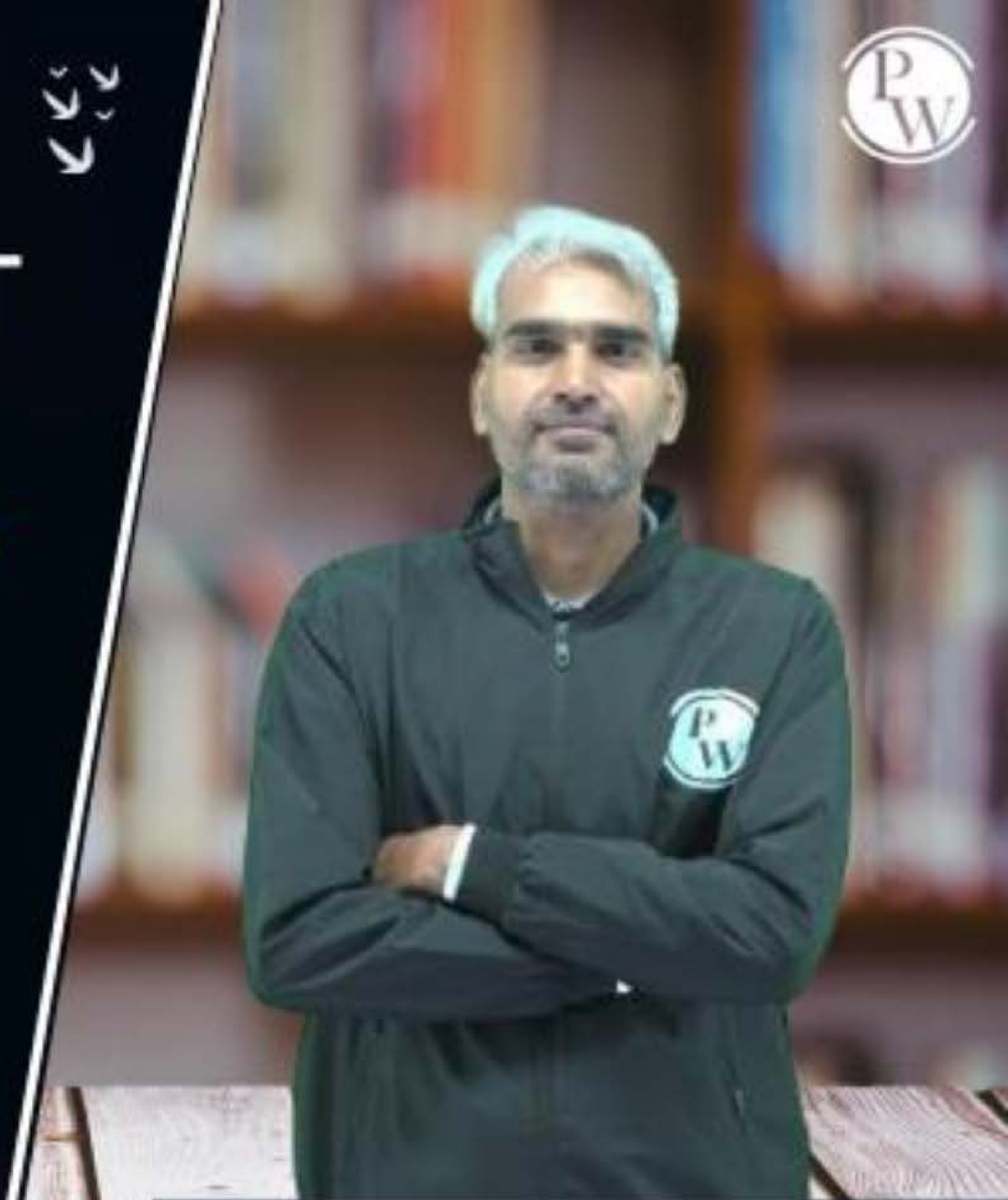
Computer Science & IT

Database Management System

Relational Model & Normal Forms

Lecture No. 01



By- Vishal Sir

Topics to be Covered

- **Topic** Syllabus
- **Topic** Introduction to DBMS
- **Topic** Relational Database Model

Slide

Topic : Syllabus

- Relational Model & Normal Forms
- Query Languages
- Transactions and Concurrency Control

Not required for DA paper

- File Organization & Indexing
- ER model

Topic : Introduction to DBMS

Data {facts/raw data}
{eg. 200 , 250
300 , 400}

Information {data with meaning}

200 is the no. of students in Vijay Batch
250 — ,, —— ,, —— Parakram — ,, —
300 — ,, —— ,, —— Shreshtha batch
400 — ,, —— ,, —— 1500 batch

database {Organized Collection of Related information}

DBMS {It is a software used to manage and access the database efficiently}

Information can be organized in many ways, and based on that various database models are defined

eg. Relational Model
Flat file model
etc

In Relational model information is organized in the form of table

In this course we will mainly discuss about Relational Model

* If we store the information gathered in previous slide in the form of relational model, then it will be of the form

Batch_name	No_of_students
Vijay	200
Parakram	250
Shreshtha	300
Super1500	400

* File system can be used to manage access the database, but File system fails when database is too large.
* Database files are stored in the blocks of secondary memory
* <u>IO Cost</u> :- IO Cost for an access of record can be defined as number of secondary memory blocks we may need to transfer to main memory in order to access that record.

Student.txt file (let size = 500 GB)

	Sid	Sname	Branch	------
B_1	S_1 S_6			
B_2	S_8 S_3			
B_3	S_2 S_7			
B_4	S_5 S_4			
⋮	⋮	⋮	⋮	⋮
B_N				

other information stored in file

Let 'N' disk blocks are required to store Student.txt file

Query: Retrieve Record of the student with Sid = S100

Manually {Time taking Process}

Write a Program

In order to write a Program we need to know the Physical storage details of the file

{it may be Security issue}

Student.txt file (let size = 500 GB)

	Sid	Sname	Branch	- - - - - -
B_1	S_1 S_6			
B_2	S_8 S_3			
B_3	S_2 S_7			
B_4	S_5 S_4			
⋮	⋮	⋮	⋮	⋮
B_N				

other information stored in file

Let 'N' disk blocks are required to store Student.txt file

Query: Retrieve Record of the Student with Sid = S100

Even if we write a program, in order to search for the record with Sid = S100 we will have to transfer the blocks of the disk containing the records of the file to main memory

Unit of transfer b/w secondary mem and main memory is '1' block

i.e at a time only one block can be transferred b/w M.M. & Sec. mem

In worst case we may have to transfer all the blocks of the file from sec. mem to main memory in order to search for a record

File System vs DBMS

File System	DBMS
(1) In order to write a program to access the records from the file we need to share the Physical details of the file with the user	(1) In DBMS, 3-tier architecture will provide (i) Data abstraction: {It hides Physical information of file from user} & (ii) Data independence → Changes performed at lower level does not affect higher levels.
(2) High IO Cost.	(2) IO Cost can be reduced in the DBMS using Indexing
(3) Low level of Concurrency {Because locking in file system is at file level, ∴ Two users (R+W/W+W) can not access the file Concurrently}	(3) High level of Concurrency (High level of Concurrency is achieved by defining locks at record level)

Topic : Relational database

{Concept of Relational model is introduced by Dr Codd in 1970}

- In relational model database is organized in the form of table {Collection of Rows & Columns}
- Each table of relational database is called a relation
- Dr. Codd defined 13 rules (numbered by 0 to 12) for a table to be defined as relation.

Topic : Relational database

eg:

Student

Attributes/fields

Sid	Sname	Branch
S_1	A	CS
S_2	A	CS
S_3	B	IT
S_4	C	CS

Records/tuples

* **Degree / Arity :-** No. of attributes in a relation is defined as degree/Arity of that relation

* **Record / Tuple :-** Each row of relation is called record/tuple

* **Cardinality :-** Number of tuples/records in a relation is defined as Cardinality of that Relation

* **Relational Schema:** Relational Schema provides the abstract details of the relation.

  eg. Name-of-Relation (Name of 1st Attribute, Name of 2nd attribute, ...)

* **Relational Instance:** If records are present in the relation then that set of records is called relational instance of that time.

Topic : Relational database

Attributes / fields

eg:

Student

Sid	Sname	Branch
S_1	A	CS
S_2	A	CS
S_3	B	IT
S_4	C	CS

$(a, b) \neq (b, a)$

- Degree = 3 (No. of Columns)
- Cardinality = 4 (No. of rows)
- Relational Schema:

  Student (Sid, Sname, Branch)

  $\neq$ Student (Sname, Sid, Branch)

Relational instance

$= \{ (S_1, A, CS), (S_2, A, CS), (S_3, B, IT), (S_4, C, CS) \}$

2 mins Summary

Topic Introduction to DBMS

Topic Relational database

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THANK - YOU

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